

Evolving Product Quality and Local Transitivity in Price Measurement

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May 2026

Abstract

This paper illustrates how chained price indices trade off updating quality standards against maintaining transitivity. Updating quality standards frequently allows price indices to reflect ongoing product innovation, but introduces path dependence, whereby measured inflation depends on the sequence of comparisons used to construct an index. Fixing quality standards ensures transitivity but limits the ability of an index to incorporate new information about quality growth. I develop a locally transitive geometric quality-adjusted unit value index (GQAUUV) that provides a structured way to characterize and reconcile this tradeoff. The key insight is that standard bilateral indices implicitly redefine quality standards in each comparison, causing standards to drift over time. When these comparisons are chained, the evolving quality standard generates path dependence in measured inflation. In contrast, the proposed index proceeds in two stages: it first constructs quality-adjusted price changes that are transitive within a pre-specified window of time, and then aggregates these comparisons across overlapping windows, effectively averaging across alternative quality standards used to measure inflation. Using data on the U.S. automobile market, I show that the locally transitive GQAUUV yields higher measured inflation as transitivity is enforced over longer time horizons. These results suggest that differences across price indices can be understood as reflecting alternative choices about how and when quality standards are updated, providing a unified perspective on the tradeoff between transitivity and timely quality adjustment.

*I am grateful to John Haltiwanger, John Shea, and Jose Asturias for their guidance and thoughtful suggestions. This paper is a preliminary draft based on the third chapter of my dissertation, *Essays in Macroeconomics*, completed at the University of Maryland. Comments are welcome. Please do not circulate without permission.

1 Introduction

Measuring the rate of inflation requires tracking the evolution of prices in the presence of ongoing product turnover and quality upgrading. As low-quality goods exit and higher-quality goods enter the market, the set of goods that consumers purchase changes continuously. To accurately capture the associated changes in inflation due to consumer substitution and product turnover, statistical agencies increasingly update processes to produce a more timely measure of inflation. For example, the Bureau of Labor Statistics now updates weights in the Chained Consumer Price Index (C-CPI) on a bi-annual basis (Reed and Perrins, 2024), and several statistical agencies have incorporated using high-frequency point-of-sale data for price measurement in particular areas of the retail sector (National Academies of Sciences et al., 2022).

While such practices improve the timeliness of price measurement, they also introduce new challenges for the construction of consistent price indices. A central tension arises because price indices constructed from sequential bilateral comparisons can be sensitive to the order in which comparisons are made, resulting in an empirical drift in the chained index relative to the direct comparison over the same time period. Multilateral index methods such as GEKS and its rolling variants (e.g. RYGEKS) are commonly used to address the associated concerns of chain drift (Ivancic et al., 2011). However, multilateral indices often lack a transparent interpretation, making it difficult to understand how transitivity is enforced. In this paper, I show that the measured rate of inflation using multilateral methods depends on the relationship between the frequency with which quality standards are updated and the horizon over which transitivity is imposed.

To clarify this relationship, I first reinterpret standard bilateral indices through the lens of a quality-adjusted unit value index. In particular, commonly used geometric matched-model indices, including Laspeyres, Paasche, and Tornqvist indices, can be viewed as special cases of a geometric quality-adjusted unit value index. Under this interpretation, bilateral indices implicitly define the standard of product quality using only prices observed in each pairwise comparison. Because this quality standard is defined separately for each pair of periods, it changes immediately as indices are sequentially chained, giving rise to issues such as chain drift and illustrating how frequent updating of quality standards can undermine transitivity.

This insight leads naturally to the construction of a locally transitive geometric quality-adjusted unit value index (GQAUV) that provides a transparent construction of a multilateral index. The proposed index proceeds in two stages. In the first stage, it constructs quality-adjusted price changes that are transitive within a pre-specified window

of time, effectively holding the quality standard fixed within each window. In the second stage, it aggregates these comparisons across overlapping windows, averaging across the quality standards implied by different comparison windows. In this way, the construction fixes quality standards locally to enforce transitivity, while allowing them to evolve over longer horizons through the aggregation across windows.

This two-stage structure makes explicit the tradeoff between timely quality adjustment and maintaining transitivity. Fixing quality adjustment factors over longer horizons ensures transitivity but delays the incorporation of new information, while updating them at each comparison captures evolving product quality but eliminates transitivity. By varying the window over which transitivity is imposed, the locally transitive GQAUV characterizes how alternative choices along this tradeoff lead to different index constructions.

Applying this framework to the US automobile market reveals a clear empirical pattern in this implementation: measured inflation varies with how persistently the quality standard is maintained over time. When the quality standard is updated at each comparison, the resulting index produces relatively low measured inflation. As the quality standard is held fixed over longer horizons, measured inflation rises. In the preferred specification, the locally transitive GQAUV yields average annual inflation of 0.59 percent, compared to approximately 0.45 percent under a benchmark index that updates the quality standard at each comparison. These differences arise even though the indices are constructed from the same underlying hedonic inputs, highlighting the role of the aggregation procedure in shaping measured inflation.

This pattern reflects the central mechanism of the paper that is observable at the item-level. Fixing quality standards over a longer period of time slows the rate at which improvements in product characteristics are incorporated into the benchmark used to construct quality-adjusted prices. As the persistence of the quality standard increases, this adjustment becomes more gradual, leading to systematically lower quality-adjusted prices.

Within this framework, alternative index constructions can be interpreted through how frequently the underlying quality standard is updated over time. Indices such as rolling time-dummy and GEKS methods rely on quality standards that are updated more frequently, either across rolling windows or at each comparison, while indices that impose transitivity over longer horizons, including the locally transitive GQAUV and full-sample approaches, maintain more persistent standards and yield higher inflation. Because the hedonic Tornqvist, GEKS, and locally transitive GQAUV indices can be constructed from a common set of quality-adjusted price relatives, comparisons across these indices iso-

late the role of the aggregation procedure itself. In particular, the hedonic Tornqvist and GEKS yield average annual inflation of approximately 0.45 percent that is systematically lower than the locally transitive GQAUV. These results show that differences across price indices are not simply empirical, but arise from how each method resolves the tradeoff between timely quality adjustment and the persistence of the underlying quality standard.

Literature Review. This paper relates to several strands of the literature on price measurement, including work on quality-adjusted unit value indices, multilateral index number methods, and hedonic approaches to quality adjustment. These literatures address complementary aspects of price measurement, but typically treat quality adjustment and transitivity separately rather than as jointly determined features of price indices. In this way, the unit value literature clarifies how quality standards are defined, multilateral methods enforce transitivity across comparisons, and hedonic approaches improve the measurement of product quality.

A natural starting point is the literature on quality-adjusted unit value indices. [von Auer \(2014\)](#) shows that traditional matched-model indices, such as the Laspeyres and Paasche indices, can be expressed as special cases of generalized unit value indices in which quality adjustment factors are constructed using prices observed in adjacent period pairs. Under this representation, bilateral indices implicitly define a standard of product quality using only information from the two periods being compared. When these bilateral comparisons are chained over time, the implicit quality standard is repeatedly redefined, which can give rise to chain drift. By contrast, when quality adjustment factors are held fixed across periods, unit value indices are transitive and therefore not subject to chain drift ([Balk, 1998](#); [de Haan, 2015a](#)). However, fixing the quality standard in this way limits the ability of the index to incorporate new information, creating a trade-off between maintaining transitivity and incorporating new information about product quality over time. For example, [Byrne et al. \(2017\)](#) illustrates this tradeoff with their proposed form of quality adjustment in the context of semiconductors.

A related literature studies multilateral index number methods designed to mitigate chain drift by transitivizing indices. Methods such as GEKS and its rolling variant RYGEKS construct transitive price indices by aggregating bilateral comparisons across multiple periods ([Ivancic et al., 2011](#); [Białek, 2025](#); [Fox et al., 2026](#)). These methods enforce transitivity at the aggregation stage, but rely on bilateral inputs with frequently updated quality standards, limiting their ability to control how quality standards evolve over time. [Aizcorbe and Ripperger-Suhler \(2026\)](#) highlight this issue in the context of

high-tech goods, showing that multilateral methods such as GEKS can be affected by a co-mingling problem in which changes in expenditure shares and product quality are jointly encoded in bilateral inputs.

A third strand of the literature focuses on hedonic methods for improving the measurement of product quality, often without imposing a structure that ensures transitivity across comparisons. Recent work has developed hedonic approaches based on first-difference specifications that account for unobserved, time-varying product characteristics (Erickson and Pakes, 2011; Ehrlich et al., 2025; Haltiwanger et al., 2026, Chapter 2 of this dissertation). These methods provide a flexible framework for constructing quality-adjusted price relatives in settings with high product turnover and evolving characteristics. Transitive hedonic indices can be constructed using time-dummy and time-product-dummy regressions to control for observable differences in product characteristics, although these approaches may be sensitive to unobserved attributes and to the assumption that the implicit valuation of characteristics remains constant over time (de Haan and Krsinich, 2018).

The approach developed in this paper builds on and connects these literatures by providing a unified framework for understanding price index construction. I show that commonly used geometric matched-model indices, including the Laspeyres, Paasche, and Tornqvist indices, can be written as special cases of a geometric quality-adjusted unit value index, highlighting how implicit quality standards are defined in bilateral comparisons. Under this interpretation, chain drift arises from the sequential redefinition of these quality standards when bilateral indices are chained over time. Building on this insight, I develop a locally transitive geometric quality-adjusted unit value index in which quality adjustment factors are defined within overlapping windows rather than separately for each pair of periods. By anchoring the quality standard within a window, the proposed index provides a transparent way to jointly determine quality adjustment and transitivity, balancing the incorporation of new information on product quality with the enforcement of transitivity over time. The framework also nests existing hedonic approaches, with the hedonic Tornqvist index emerging as a special case, linking the method directly to widely used empirical implementations. In this way, the proposed framework unifies these strands of the literature by making explicit the tradeoff between updating quality standards and enforcing transitivity using modern methods in the price measurement literature.

Roadmap. The rest of this paper is structured as follows. Section 2 develops the conceptual framework. Section 3 describes the data and measurement strategy for the case of

US automobile sales. Section 4 presents the empirical results and compares the resulting price index with conventional bilateral and multilateral alternatives. Section 5 concludes.

2 Conceptual Framework

This section develops the conceptual framework for the locally transitive geometric average quality-adjusted unit value index (GQAUV). A central challenge in price measurement is that updating quality standards to reflect new information can undermine transitivity, while enforcing transitivity requires holding these standards fixed over time. I formalize this tradeoff by constructing price indices that fix quality standards within a moving window to obtain locally transitive comparisons, and then reconcile these comparisons into a single transitive price index. This two-stage structure separates the enforcement of transitivity within windows from the evolution of quality standards across windows, providing a transparent framework for understanding how alternative index constructions balance these competing objectives.

2.1 Geometric Average Quality-Adjusted Unit Value Index

The goal of this section is to construct a measure of price inflation that reflects changes in the cost of purchasing a basket of goods while accounting for time-varying product quality and product turnover. I use a geometric average quality-adjusted unit value index (GQAUV) as an estimator for these price changes based on observed prices, expenditure shares, and a quality standard, which I describe in more detail below.

The geometric quality-adjusted unit value equals the expenditure share weighted geometric average of quality-adjusted prices for a basket of goods (Ω)

$$(1) \quad P_t = \prod_{k \in \Omega_t} (p_{kt}/\lambda_{kt})^{s_{kt}},$$

and the GQAUV is the ratio of quality-adjusted average prices across periods

$$(2) \quad \text{GQAUV}_{t,t-1} = \frac{P_t}{P_{t-1}},$$

where p_{kt} and $s_{kt} = p_{kt}q_{kt}/\sum_{k \in \Omega_t} p_{kt}q_{kt}$ denote the price level and expenditure share of good k in period t for a basket Ω .¹ Additionally, λ_{kt} denotes the quality adjustment factor

¹The average quality-adjusted price using the standard quality-adjusted unit value is traditionally written as an expenditure share weighted harmonic mean of quality adjusted prices:

that accounts for price variation attributed to product heterogeneity for item k in period t . Differences in quality adjustment factors potentially capture variation in units, observable characteristics, or unobserved product attributes that affect the effective quantity delivered by a good. Importantly, alternative choices of quality adjustment factors correspond to different index number formulas (von Auer, 2014), so the definition of quality adjustment factors governs how product heterogeneity is incorporated into measured inflation as opposed to an exact representation of consumer utility (Balk, 1998).

To expand on this point, I view quality adjustment factors as item- and time-specific scaling terms that normalize observed prices relative to a given quality standard. More formally, a quality standard can be defined as a rule that maps observable data into quality adjustment factors in a systematic fashion,

$$\lambda_{kt} = Q(X \mid I, \theta),$$

where Q denotes the transformation associated with the quality standard, X denotes the set of available data, I denotes the reference period, and θ denotes parameters governing this transformation. Quality standards are therefore defined by the joint set $\{Q, I, X, \theta\}$, and quality adjustment factors are the resulting outputs induced by this system.

To ground ideas, matched-model price indices implicitly implement quality standards that transform available data, primarily price levels, into adjustment factors (von Auer, 2014). For example, consider two Laspeyres price indices constructed over different comparison windows. A Laspeyres index estimates the average price change between periods $t - j$ and t using prices in period t as quality adjustment factors, so that $\lambda_{k\tau_1} = p_{kt}$ for $\tau_1 \in \{t - j, t\}$. By contrast, a Laspeyres index comparing periods t and $t + j$ uses prices in period $t + j$ as quality adjustment factors, so that $\lambda_{k\tau_2} = p_{kt+j}$ for $\tau_2 \in \{t, t + j\}$. In both cases, the Laspeyres index uses current-period prices to normalize price variation, but the underlying quality standard shifts as the reference period changes, even though the index formula remains the same.

This distinction between quality standards and quality adjustment factors clarifies the role of quality adjustment in price index construction. While existing approaches often interpret adjustment factors as reflecting underlying utility or product quality, I instead treat them as the outcome of a chosen standard of comparison. Under this interpretation, the same index number formula applied to different periods of time corresponds to differ-

$\sum_{k \in \Omega_t} p_{kt} q_{kt} / \sum_{k \in \Omega_t} \lambda_{kt} q_{kt} = \left(\sum_{k \in \Omega_t} s_{kt} [p_{kt} / \lambda_{kt}]^{-1} \right)^{-1}$. Hence, the average quality-adjusted price using a geometric quality adjusted unit value differs from the alternative formula primarily in its aggregator as opposed to its weighting scheme for example.

ent quality standards that govern how prices are normalized. Making this object explicit provides a unified framework for understanding how alternative indices incorporate new or additional information to normalize price variation.

2.2 Quality Standards and the Representation of Matched-Model Indices

Building on the distinction between quality standards and quality adjustment factors, I now focus on a class of quality standards where the GQAUV coincides with popular bilateral matched-model indices. This class of standards shows that differences across standard index number formulas arise from differences in the underlying quality standard. In particular, when quality standards are defined using different prices from the same reference periods, the GQAUV nests the geometric matched-model Laspeyres, Paasche, and Tornqvist indices. To illustrate this connection, I temporarily abstract from product turnover.²

To characterize how quality standards shape index number formulas, I consider a series of adjustment factors that depend on prices relative to a common reference level. I define a class of quality standards using the bounding approach from [Haltiwanger et al. \(2026\)](#), which generate quality adjustment factors of the form

$$(3) \quad \lambda_k^{t-1,t}(\alpha) = \left(\frac{p_{kt}}{P_t} \right)^\alpha \left(\frac{p_{kt-1}}{P_{t-1}} \right)^{1-\alpha} \quad \text{where } \alpha \in [0, 1].$$

In terms of the framework introduced above, this corresponds to a particular choice of quality standards that construct adjustment factors using a convex combination of relative price dispersion around the average quality-adjusted price, with the normalization by P_t ensuring scale invariance and the mixing parameter α governing the extent to which the quality standard reflects the cross-sectional distribution of prices in the current and base periods.³ Using this class of quality standards, the GQAUV nests the set of standard

²I revisit the issues associated with product turnover at the end of this section. I discuss the empirical challenges associated with accounting for product turnover in the next section after discussing the available data and measurement details. Both features are incorporated in the general framework developed in Chapter 2 for the case of quality-adjusted unit value indices constructed on a bilateral basis.

³In practice, aggregate price terms are omitted from quality adjustment factors because they cancel in index number construction. Average quality-adjusted prices are calculated after defining adjustment factors, so their inclusion in λ_k is not required when calculating the rate of inflation.

matched-model indices depending on the choice of the mixing parameter:

$$\frac{P_t^{t-1,t}(\alpha)}{P_{t-1}^{t-1,t}(\alpha)} = \begin{cases} \prod_{k \in \Omega} (p_{kt}/p_{kt-1})^{s_{kt-1}} & \text{if } \alpha = 1 \\ \prod_{k \in \Omega} (p_{kt}/p_{kt-1})^{0.5(s_{kt-1}+s_{kt})} & \text{if } \alpha = 0.5 \\ \prod_{k \in \Omega} (p_{kt}/p_{kt-1})^{s_{kt}} & \text{if } \alpha = 0 \end{cases}$$

where $P_t^{t-1,t}(\alpha)$ denotes the geometric average quality-adjusted price of goods in period t constructed using adjustment factors based on prices observed in periods $t-1$ and t with mixing parameter α . Therefore, the GQAUUV using the bounding approach equals the geometric Laspeyres and Paasche indices at the extreme values of the mixing parameter, and at the midpoint value $\alpha = 0.5$, the GQAUUV equals the Tornqvist price index, illustrating how different index formulas arise from alternative specifications of the quality standard.

2.3 Extending Quality Standards Over Time

While the GQAUUV nests standard bilateral indices under the bounding approach, it will not yield a transitive price index when comparisons are chained over time.⁴ The source of this failure is that the quality standard's reference period is specific to each bilateral comparison. As a result, chaining adjacent comparisons involves repeated re-basing of the underlying quality standard, so that the resulting index may exhibit drift in measured inflation.⁵

To formalize this point, consider chaining adjacent bilateral GQAUUV comparisons between periods $t-1$ and $t+1$. In general, chaining does not reproduce the direct comparison across multiple periods as the quality-adjusted average price in period t changes across intermediate comparisons

$$P_t^{t,t+1}(\alpha) \neq P_t^{t-1,t}(\alpha).$$

⁴For reference, transitivity in the price measurement literature implies inflation rates exhibit the transitive property. For example, a transitive rate of inflation calculated between periods $t-1$ and $t+1$, denoted as $\mathbb{P}_{t-1,t+1}$, is equivalent to the product of intermediate rates: $\mathbb{P}_{t-1,t+1} = \mathbb{P}_{t-1,t} \times \mathbb{P}_{t,t+1}$. In the literature, quality-adjusted unit value indices are seen as transitive under a general definition of quality adjustment factors (de Haan and Krsinich, 2018).

⁵This drift can arise from classical chain drift, reflecting a correlation between price relatives and index weights, or from a co-mingling problem, in which product quality is evaluated relative to different standards across periods (see Aizcorbe and Ripperger-Suhler, 2026).

Similar to the example provided above with the Laspeyres price index, this failure of transitivity reflects the fact that quality adjustment factors are rebased across adjacent periods (i.e. $\lambda_k^{t,t+1}(\alpha) \neq \lambda_k^{t-1,t}(\alpha)$) as each comparison is constructed relative to a different quality standard.

A natural way to restore transitivity is therefore to construct price indices that hold the quality standard fixed over multiple periods. I implement this idea by defining quality standards in a similar fashion to the bounding approach but over a longer window of time. This construction yields a locally transitive price index, in the sense that comparisons within the window are consistent with one another.

To illustrate, consider a simple example between periods $t - 1$ and $t + 1$. I define quality standards using a convex combination of prices from the first and last periods in this window of time

$$\lambda_k^{t-1,t+1}(\alpha) = \left(\frac{p_{kt+1}}{P_{t+1}} \right)^\alpha \left(\frac{p_{kt-1}}{P_{t-1}} \right)^{1-\alpha}.$$

which I will refer to as the extended bounding approach as it defines adjustment factors over a longer time horizon than adjacent periods. Under this specified form of quality adjustment, the quality-adjusted average price in period t is constant as the quality standard does not change across comparisons. As a result, the GQAUUV is locally transitive across the three periods

$$\frac{P_{t+1}^{t-1,t+1}(\alpha)}{P_{t-1}^{t-1,t+1}(\alpha)} = \frac{P_{t+1}^{t-1,t+1}(\alpha)}{P_t^{t-1,t+1}(\alpha)} \times \frac{P_t^{t-1,t+1}(\alpha)}{P_{t-1}^{t-1,t+1}(\alpha)}$$

Moreover, chaining adjacent comparisons yields the same result as the direct comparison, which again equates to the corresponding direct matched-model indices defined over the same horizon

$$\frac{P_{t+1}^{t-1,t+1}(\alpha)}{P_{t-1}^{t-1,t+1}(\alpha)} = \begin{cases} \prod_{k \in \Omega} (p_{kt+1}/p_{kt-1})^{s_{kt-1}} & \text{for } \alpha = 1 \\ \prod_{k \in \Omega} (p_{kt+1}/p_{kt-1})^{0.5(s_{kt-1} + s_{kt+1})} & \text{for } \alpha = 0.5 \\ \prod_{k \in \Omega} (p_{kt+1}/p_{kt-1})^{s_{kt+1}} & \text{for } \alpha = 0 \end{cases}$$

where specific values of the mixing parameter determines whether the GQAUUV equates to the direct geo-Laspeyres, Tornqvist, and geo-Paasche between periods $t - 1$ and $t + 1$.

The key insight from extending the bounding approach is that transitivity is enforced by holding the underlying quality standard fixed over time. While direct comparisons

summarize price changes over multiple periods, they do not specify how those changes are distributed across intermediate periods. By fixing the quality standard within the window, the GQAUV recovers a sequence of intermediate price changes that is consistent with the corresponding direct comparison, ensuring that all comparisons are evaluated relative to a common standard. This construction therefore provides a locally transitive decomposition of multi-period price changes into a path of quality-adjusted intermediate price changes. At the same time, holding the quality standard fixed within the window delays the incorporation of new information about product quality, illustrating the trade-off between maintaining transitivity and updating quality standards over time.⁶

2.4 Multiple Estimates for the Same Rate of Inflation

Because the choice of window length determines the reference periods used to define quality adjustment factors, the resulting GQAUV depends on the choice of window length and mixing parameter. Different windows therefore generate distinct, locally transitive estimates of inflation for the same pair of periods even when the mixing parameter is fixed. For example, a GQAUV constructed using the extended bounding approach over an arbitrary window $[t - W, t]$ with mixing parameter equal to 0.5 coincides with the direct Tornqvist index when chained over the full window, but its implied intermediate price change between periods $t - 1$ and t will generally differ from the Tornqvist index computed directly over those adjacent periods.⁷ In other words, there are multiple estimates for a single rate of inflation that differ and must be reconciled.

I formalize these discrepancies in a general manner. In doing so, I focus on the rate of inflation between an arbitrary base period $t - j$ and the current period t using window lengths W_0 and W_1 such that $j \leq \min\{W_0, W_1\}$. Both rates are constructed using the GQAUV with the extended bounding approach. However, because the window lengths differ, these two rates in general do not equal

$$\frac{P_t^{t-W_0,t}(\alpha)}{P_{t-j}^{t-W_0,t}(\alpha)} \neq \frac{P_t^{t-W_1,t}(\alpha)}{P_{t-j}^{t-W_1,t}(\alpha)}$$

Moreover, neither rate is ex-ante more informative about the rate of inflation. They simply

⁶This logic extends directly to longer horizons as well. For any window of N consecutive periods, define quality adjustment factors using prices from the endpoints of the window: $\lambda_k^{t-N,t}(\alpha) = p_{kt}^\alpha p_{kt-N}^{1-\alpha}$. This construction ensures local transitivity as all comparisons within the window are evaluated relative to a common quality standard.

⁷For completeness, I am implicitly assuming that $W \geq 1$, so the GQAUV measures a rate of inflation that is over a longer horizon than the Tornqvist between periods $t - 1$ and t .

trade off how long an index remains transitive versus the relevancy of the underlying quality standards. This observation motivates the need for a second-stage aggregation procedure that reconciles these locally transitive but mutually inconsistent estimates into a single price index.

2.5 Second-Stage Aggregation

I utilize a second-stage aggregation method, similar to the cycle method, to reconcile the varying estimates that can be produced using the GQAUUV with the extended bounding approach.⁸ This method of aggregation can be interpreted as solving for a single transitive price by aggregating across overlapping comparisons that differ in terms of their underlying quality standards.

General Set-Up. I define the transitive price index associated with the mixing parameter α as the sequence of log price levels $\{\ln P_t(\alpha)\}_t$ that best reconciles the set of locally transitive comparisons constructed across window lengths. Let $\Delta \ln P(\alpha, W)$ denote the vector of log average price changes constructed using the GQAUUV with the extended bounding approach across all available windows and period pairs, and let A denote the difference matrix that maps log price levels to log price changes.

I recover the transitive price index, which I refer to as the locally transitive GQAUUV, by solving the least-squares problem

$$(4) \quad \widetilde{\ln P(\alpha)} = \arg \min \|A \ln P(\alpha) - \Delta \ln P(\alpha, W)\|^2.$$

This criterion selects the sequence of price levels that is most consistent, in a least-squares sense, with the set of locally transitive comparisons. Because price levels are identified only up to a normalization, I impose $\widetilde{\ln P_{t-W}(\alpha)} = 0$, so that all price levels are measured relative to a base period.

To interpret this estimator, it is useful to represent the window-specific comparisons as

$$\Delta \ln P_{t-j,t}^{t-W,t}(\alpha) = \widetilde{\ln P_t(\alpha)} - \widetilde{\ln P_{t-j}(\alpha)} + \varepsilon_{t-j,t}^{t-W,t}(\alpha),$$

where $\varepsilon_{t-j,t}^{t-W,t}(\alpha)$ captures discrepancies across comparisons arising from differences in the

⁸See Willenborg (2018) for more information on the cycle method. The second-stage aggregation is analogous to multilateral cycle methods, but differs in that it reconciles discrepancies across locally transitive comparisons constructed under alternative quality standards. Consequently, the aggregation procedure operates over a structured set of inputs that differ systematically in terms of their quality standards, rather than over a set of bilateral comparisons that are intransitive.

underlying quality standards. Under this representation, the least-squares estimator in equation (4) selects the transitive price system that minimizes the aggregate squared discrepancies across all comparisons. Importantly, these discrepancies do not reflect classical measurement error, but rather systematic differences induced by the use of alternative quality standards across windows. As a result, the estimator can be interpreted as reconciling a set of locally transitive but mutually inconsistent comparisons into a single transitive price index.

Additional Considerations. In practice, there are additional choices that can be made to construct the locally transitive GQAUUV and are discussed later with respect to the paper’s empirical application and results. To briefly highlight them here, these include estimating the price index on a rolling basis, sample selection and the use of sample weights, and the selection of window lengths. With respect to a rolling estimate, this procedure can be implemented sequentially. At each period t , comparisons constructed within the longest window $[t - W, t]$ are added to the system, while older comparisons are removed as the window advances. Solving equation (4) recursively yields a rolling estimate of the price index that remains consistent with the locally transitive comparisons available at each point in time. More generally, alternative weighting schemes and choices of the maximum window length determine how different comparisons contribute to the resulting index, but do not alter the underlying aggregation framework described above.

In the empirical analysis, I illustrate how sample weights can be applied, for example, to place equal weight on inflation rates or on windows. I also suggest using the empirical context to guide the choice of the maximum window length. For instance, a monthly index may call for a maximum window length of 13 months, in line with common implementations of the RYGEKS index. In the application below, I use annual data and select the maximum window length based on observed product life cycles.

Aggregation Example. To ground ideas, I will set-up this approach using a three period case spanning periods 0 to 2 for simplicity. I will use rates of inflation that come from one of three windows that are a subset of the window $[0, 2]$: $[0, 2]$, $[0, 1]$, and $[1, 2]$. Given the set of available windows, there are five rates of inflation that can be calculated directly from the data. All rates are calculated using the same mixing parameter α , so to minimize notation, I suppress the parameter from the index. For illustrative purposes, I suppose

the following rates come from data

$$\begin{bmatrix} \Delta \ln P_{1,2}^{0,2} \\ \Delta \ln P_{0,1}^{0,2} \\ \Delta \ln P_{0,2}^{0,2} \\ \Delta \ln P_{0,1}^{0,1} \\ \Delta \ln P_{1,2}^{1,2} \end{bmatrix} = \begin{bmatrix} 0.10 \\ 0.07 \\ 0.17 \\ 0.08 \\ 0.08 \end{bmatrix}$$

Moreover, there are three aggregate price levels in the corresponding index: $\ln P_0$, $\ln P_1$, and $\ln P_2$. Given this information, I can set up a system of linear equations

$$\begin{bmatrix} 0.10 \\ 0.07 \\ 0.17 \\ 0.08 \\ 0.08 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} \ln P_2 \\ \ln P_1 \\ \ln P_0 \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,2}^{0,2} \\ \varepsilon_{0,1}^{0,2} \\ \varepsilon_{0,2}^{0,2} \\ \varepsilon_{0,1}^{0,1} \\ \varepsilon_{1,2}^{1,2} \end{bmatrix}$$

which provides a number of potential points of discussion. First, the equation provides an intuitive illustration showing how the A matrix transforms price levels into rates of inflation. Second, it can be seen from this example that the A matrix has two linearly independent column vectors, meaning it is rank deficient and the price index must be normalized. In particular, I will normalize the index such that $\ln P_0 = 0$, implying that the values for $\ln P_1$ and $\ln P_2$ are relative to the base period. Lastly, because there are more equations than unknowns, this system is over-identified. In particular, several rows correspond to multiple estimates of the same rate of inflation constructed under different window lengths. For example, the first and last rows provide distinct estimates of inflation between periods 1 and 2, reflecting different underlying quality standards. The second-stage aggregation reconciles these discrepancies by selecting the set of price levels that is most consistent with the full set of comparisons.

With this representation in place, the price index can be recovered by solving for the set of price levels that is consistent with the system, given the normalization $\ln P_0 = 0$. Using equation (4) as the corresponding objective function, this example yields the following values:

$$\widehat{\ln P_1} = 0.076, \quad \widehat{\ln P_2} = 0.168.$$

which imply the following reconciled rates of inflation:

$$\widehat{\Delta \ln P_{0,1}} = 0.076, \quad \widehat{\Delta \ln P_{1,2}} = 0.092, \quad \widehat{\Delta \ln P_{0,2}} = 0.168,$$

By construction, the rates of inflation satisfy transitivity. The reconciled price index also differs from the original inputs because the estimates are not mutually consistent. The second-stage aggregation selects price levels that balance these competing estimates. For example, the implied rate of inflation between periods 1 and 2 lies between the two input values of 0.10 and 0.08, while the longer-horizon comparison between 0 and 2 is adjusted downward from 0.17 to 0.168. The resulting system provides a single transitive price path that is most consistent with the set of locally transitive comparisons.

Comparison to Alternative Multilateral Indices. It is useful to contrast this approach with multilateral index methods such as the GEKS index. Using the notation introduced in this paper and focusing on the Tornqvist index, which corresponds to the GQAUUV with a mixing parameter $\alpha = 0.5$, a rolling GEKS (RGEKS) index from periods $t - 1$ to t can be written as follows

$$\text{RGEKS}_{t-1,t} = \prod_{w=1}^W \left[\frac{P_{t-w}^{t-w,t-1}(0.5)}{P_{t-1}^{t-w,t-1}(0.5)} \times \frac{P_t^{t-w,t}(0.5)}{P_{t-w}^{t-w,t}(0.5)} \right]^{\frac{1}{W}}.$$

This construction enforces transitivity by averaging across alternative price paths that connect periods $t - 1$ and t through intermediate periods $t - w$ for $1 \leq w \leq W$.

From the perspective developed in this paper, these price paths are constructed from bilateral indices that rely on different quality standards, as each input uses a distinct set of reference periods. As a result, the RGEKS index reconciles discrepancies across paths that are themselves based on incompatible standards of comparison. In contrast, the locally transitive GQAUUV holds the quality standard fixed within each window, ensuring that all within-window comparisons are evaluated relative to a common standard. The second-stage aggregation then reconciles discrepancies across these competing standards, rather than across alternative price paths. This distinction highlights a key difference between the two approaches.

Finally, [Willenborg \(2018\)](#) shows that the RGEKS index can be interpreted as a special case of the cycle method, which relies on a least squares solution. From this perspective, the RGEKS index can be viewed as averaging over a selected set of paths connecting the relevant periods. However, this set represents only a subset of all possible comparisons implied by the data, so the resulting index reflects an average over a restricted portion of

the space of potential inflation paths. In contrast, the locally transitive GQAUV places no restrictions on the set of admissible comparisons in the second-stage aggregation, thereby allowing for a more general characterization of inflation by incorporating a broader set of internally consistent comparisons.

2.6 Incorporating Product Turnover

The conceptual framework developed above abstracts from product turnover to highlight the role of quality standards in generating tradeoffs between transitivity and timely quality adjustment. In practice, however, product entry and exit are pervasive, particularly in disaggregated data, and introduce an additional challenge for price measurement.

With product turnover, the set of observed goods differs across periods, so that prices are not available for all items in each period. As a result, quality adjustment factors cannot be constructed directly for all goods using observed data alone, even when the quality standard is fixed. This issue arises because the data entering the quality standard, X , is incomplete, preventing a consistent application of the quality standard $Q(X | I, \theta)$ across comparisons.

A common approach in the literature to overcome the lack of sufficient price information is to impute it using hedonics, so that all goods can be evaluated relative to a common quality standard once specified. In this paper, I adopt a full-imputation approach based on hedonic methods, which constructs predicted prices for missing observations using observable product characteristics. This procedure effectively augments the dataset to a complete panel, allowing quality adjustment factors to be constructed for all goods across all relevant periods.

However, the empirical application introduces its own challenges and choices. Because missing prices are imputed rather than observed, the resulting index depends on the specification of the hedonic model used to construct predicted prices. In particular, the distinction between price change and quality change becomes conditional on the assumed relationship between observable characteristics and prices. These issues are discussed in more detail in the subsequent section.

3 Empirical Application

This section describes the data and measurement strategy used to construct the locally transitive GQAUV. I focus on measuring inflation in the US market for new automobiles, where detailed data on prices, quantities, and product characteristics allow for explicit

construction of the locally transitive GQAUUV in the presence of product turnover. The empirical application is intended to be exploratory, illustrating how the framework can be implemented in practice under a particular set of modeling and measurement choices.

The analysis proceeds in three steps. First, I describe the automobile data and construct product-level measures of prices and quantities, which provide the empirical foundation for the analysis. Second, I estimate a hedonic pricing model to construct predicted price relatives that complete the set of price comparisons in the presence of product turnover. Third, I use these predicted price relatives to construct quality adjustment factors and aggregate across alternative rates of inflation to obtain the locally transitive GQAUUV.

3.1 US Market of Automobiles

For the empirical analysis, I construct price indices for the US automobile market from 1980 to 2018. The data are drawn from the replication package of [Errico and Lashkari \(2025\)](#), originally compiled from the Wards Automotive Yearbook. The dataset provides annual information on model-level prices, sales, and product characteristics, including horsepower, fuel economy, weight, dimensions, and body style. I use Manufacturer's Suggested Retail Price (MSRP) as the price measure and define quantities as sales divided by MSRP, which proxies for physical units in the absence of direct quantity measures.

A key feature of this empirical setting is the combination of rich product characteristics and substantial product turnover. Table 1 reports average bilateral turnover rates across adjacent years. While the majority of products are present in consecutive periods, nearly 19 percent of goods either enter or exit in a given year. This degree of turnover limits the applicability of matched-model approaches and complicates quality adjustment when comparisons span multiple periods.

These features make the automobile market a useful environment for evaluating alternative approaches to quality adjustment. The availability of detailed characteristics allows for the construction of comparable price measures across heterogeneous products, while the presence of substantial turnover implies that many product-level price comparisons are not directly observed. As a result, the data required to evaluate price changes consistently across periods are incomplete when relying on observed prices alone. In this setting, hedonic methods provide a way to construct the missing comparisons needed to place products on a common basis.

Table 1: Annual Turnover Rates in the US Market

	Mean (1)	SD (2)	Min (3)	Max (4)
Continuing	0.814	0.097	0.587	0.912
Entering	0.101	0.053	0.025	0.218
Exiting	0.086	0.048	0.034	0.197

Notes: The table reports bilateral turnover rates for automobiles in the US market from 1980 to 2018. For each pair of adjacent years, products are classified as continuing (present in both years), entering (present only in the current year), or exiting (present only in the base year). Entries report the share of products in each set, averaged across all adjacent-year pairs. Shares sum to one within each year pair.

3.2 Estimation of the Hedonic Pricing Model

This subsection describes the estimation of the hedonic model used to construct predicted price relatives. These predicted price relatives form the key input into the quality adjustment procedure described in the following subsection. I primarily follow [Pakes \(2003\)](#), which emphasizes that product turnover induces a selection problem in matched-model comparisons, as goods that exit are systematically associated with declining prices and therefore omitted from observed price changes. A central implication of this framework is that the hedonic price function should be allowed to vary over time to reflect changing valuations of product characteristics and market conditions. In practice, this motivates estimating hedonic regressions separately for each period, so that predicted price relatives are constructed using time-varying mappings between characteristics and prices.

The procedure follows the [Erickson and Pakes \(2011\)](#) (EP-TV) framework and allows unobserved product characteristics to evolve over time while maintaining a stable mapping between observed characteristics and prices. Conceptually, the model combines a cross-sectional relationship between price relatives and characteristics with a dynamic component that allows unobserved product quality to evolve over time. The empirical implementation follows a two-stage procedure that first recovers a measure of unobserved quality and then incorporates this information into price changes. I begin by estimating a log-level hedonic model in the base period:

$$\ln p_{kt-1} = X_k' \beta_{t-1} + e_{kt-1},$$

where X_k denotes a vector of observed product characteristics. The estimated residual $\hat{e}_{kt-1}$ captures unobserved product characteristics. I then incorporate this residual into a

Table 2: Summary Statistics of Product Characteristics

	Mean (1)	SD (2)	Min (3)	Max (4)
Width (inches)	71.0	1.028	67.9	73.6
Length (inches)	185	1.015	180	189
Weight (pounds)	3,390	1.119	2,890	3,890
Horsepower	167	1.352	98.8	234
Miles per Dollar	8.46	1.307	4.95	12.70
US Manufacturer	0.459	0.101	0.282	0.616
European Manufacturer	0.198	0.052	0.124	0.291
Luxury Model	0.292	0.058	0.209	0.416
Sports Model	0.109	0.015	0.087	0.138
Truck	0.074	0.034	0.031	0.189
SUV	0.207	0.120	0.033	0.376
Van	0.067	0.016	0.039	0.099

Notes: This table reports summary statistics for the product characteristics used in the hedonic model. Continuous characteristics are measured in levels, while indicator variables denote product origin and model class. Summary statistics are computed across all product-year observations in the sample.

log-difference specification:

$$\Delta \ln p_{kt} = X'_k \eta_t + \kappa_t \hat{e}_{kt-1} + u_{kt}$$

where the vector of coefficients $\{\hat{\eta}_t, \kappa_t\}$ are used to impute predicted price relatives, as is discussed in detail in the next subsection.

To maintain a consistent approach to estimating the hedonic model, I fix the set of observable product characteristics used as inputs. Table 2 reports summary statistics for the set of product characteristics used in estimation. The specification includes measures of vehicle size, performance, fuel efficiency, origin, and model class. Continuous characteristics such as width, length, weight, horsepower, and miles-per-dollar enter in logs. These variables capture the key observable dimensions along which automobiles differ and provide sufficient variation to identify the relationship between characteristics and prices.

While these characteristics account for observable differences across goods, they do not span the full characteristics space. The first-stage residual from the EP-TV framework therefore serves as a proxy for latent product characteristics. By including this residual in the second-stage regression for continuing and exiting, the EP-TV framework allows

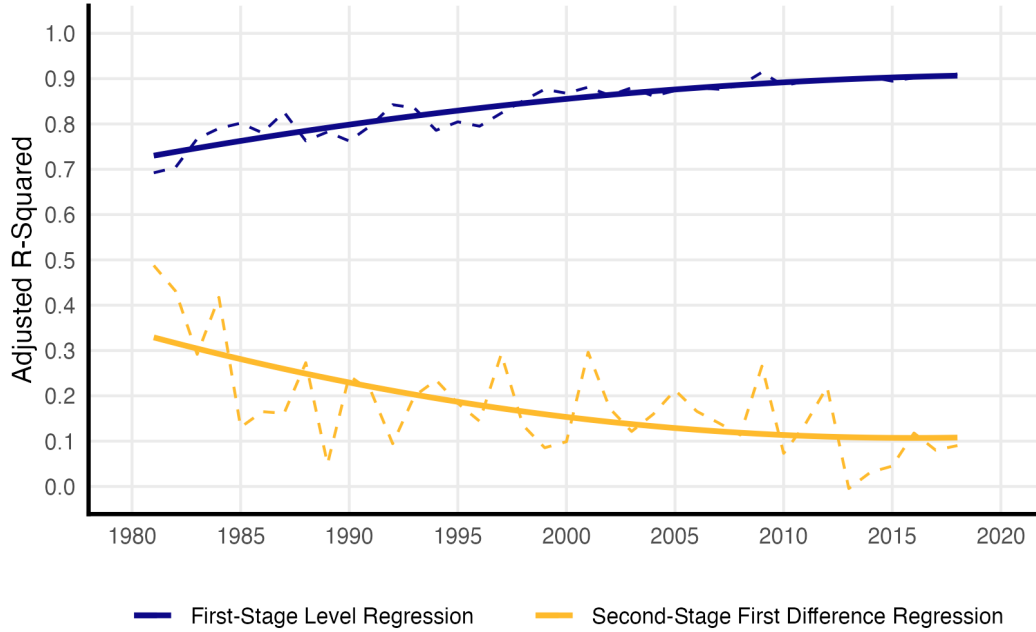


Figure 1: Adjusted R-Squared from the Two-Stage Hedonic Model

Notes: The figure reports the adjusted R-squared from the first and second stage of the hedonic model run for each pair of adjacent periods. The dashed lines in the figure report the observed adjusted R-squared from each regression. The solid lines represent a quadratic polynomial fit to the observed adjusted R-squared across years.

the contribution of unobserved characteristics to vary over time. This feature is particularly important in settings with frequent product entry, exit, and redesign, where both observed and unobserved attributes evolve over time.

To construct a consistent time series of predicted price relatives, I estimate the model separately for each adjacent period pair from 1980 to 2018 using this fixed set of characteristics. This approach allows the valuation of characteristics to change over time while preserving a common specification across periods. Figure 1 reports the adjusted R^2 from the first-stage (log-level) and second-stage (log-difference) regressions. The first-stage regressions exhibit high and increasing explanatory power, while the second-stage regressions display substantially lower and declining explanatory power, with adjusted R^2 values between 0.1 and 0.3. These patterns are consistent with [Erickson and Pakes \(2011\)](#) and reflect a tradeoff inherent in the specification: a fixed set of characteristics provides a stable and interpretable mapping for price levels but captures only a limited share of variation in price changes. Despite this limitation, the model delivers a consistent set of predicted price relatives that can be used to construct a characteristics-based price index.

3.3 Construction of Predicted Price Relatives

This subsection describes how predicted price relatives are constructed using the estimated hedonic models. The two-stage EP-TV procedure provides a mapping from product characteristics to price relatives on an annual basis. In particular, the estimated coefficients define a relationship between observed characteristics and price changes that can be used to construct predicted price relatives between periods $t - 1$ and t .

To apply this mapping in the multilateral setting, the remaining task is to define a characteristic vector for each good across the relevant time horizon so that the hedonic price functions can be evaluated in a consistent manner. While the EP-TV residual accounts for unobserved product characteristics, the construction of predicted price relatives requires a complete set of observable characteristics for each good in each period. However, product characteristics are time-varying due to vehicle redesigns and only recorded in periods with positive sales. Therefore, they are not observed for all goods across all periods due to both true and temporary product turnover.

To address this issue, I assign observable characteristics using the nearest available observation in time for each product. Specifically, for each vehicle k in period t , I identify the period s with available data that minimizes $|s - t|$, and use the corresponding characteristics as inputs to the hedonic model.⁹ This procedure defines a complete characteristic space over which the time-varying hedonic price functions can be evaluated, while allowing for gradual changes in product attributes over time. In the sample, 7.8 percent of vehicles experience temporary product turnover.

In addition to observable characteristics, the construction of predicted price relatives also requires values of the first-stage residual. By construction, the residual is recovered only for goods observed in the base period $t - 1$. For goods that are not observed in period $t - 1$, the residual is not available and must be imputed. Following [Ehrlich et al. \(2025\)](#), I set the residual equal to zero for goods that enter in period t , that is, goods observed in period t but not in period $t - 1$.¹⁰ I extend this treatment to all goods for which the residual is not observed, including those that are absent in both periods $t - 1$ and t . This imposes a symmetric rule for handling missing residuals, so that predicted price relatives for all such goods are constructed using the same mapping from observed characteristics to price changes.

⁹For item-period pairs in which observable characteristics are not recorded, I assign characteristics using the nearest available observation in time for the same vehicle. When two periods are equally close, I select the earlier period.

¹⁰In their main specification, [Ehrlich et al. \(2025\)](#) set the residual equal to zero for entering goods and find that price indices are robust to alternative choices.

Given the characteristic vectors and residual values defined above, predicted price relatives are constructed by evaluating the estimated hedonic price function for each good across periods. For goods not observed in a given period, the hedonic mapping is evaluated using the available characteristic information for that good, with characteristics assigned using the nearest available observation in time only when not directly observed.

Note that, while this procedure yields a complete set of predicted price relatives, the resulting comparisons depend on the modeling and imputation choices embedded in their construction. In particular, predicted price relatives inherit the specification of the hedonic model used to map characteristics into price changes, as well as the assumptions used to handle missing data, including the assignment of characteristics across periods and the imputation of residuals for goods not observed in the base period. In addition, constructing price relatives that span multiple periods requires a choice of how to aggregate or extend adjacent-period predictions. These alternatives reflect different trade-offs between incorporating additional information and preserving the interpretation of intermediate price movements. As a result, the resulting price comparisons should be interpreted as reflecting a particular implementation of the framework, rather than as model-free measures of price change. In this sense, the empirical application remains exploratory and is intended to illustrate how the framework can be applied in practice.

3.4 Construction of Quality Adjustment Factors

With the necessary set of predicted price relatives computed, I next construct quality adjustment factors that are used to build the GQAUV. To illustrate how these predicted price relatives enter the construction, it is useful to first express the quality standard under the extended bounding approach in terms of item-level price relatives. In particular, the quality adjustment factor for a good k in period t within the reference period from $t - j$ to $t + i$ can be written as

$$\tilde{\lambda}_{kt}^{t-j, t+i}(\alpha) = \left(\frac{p_{kt+i}}{p_{kt}} \right)^{\alpha} \left(\frac{p_{kt-j}}{p_{kt}} \right)^{1-\alpha} p_{kt},$$

which combines price relatives from periods $t - j$ and $t + i$ to period t , respectively. This representation makes clear that quality adjustment factors can be constructed from item-level price relatives, which will be replaced by predicted price relatives in the empirical implementation. Aggregate quality-adjusted price terms are omitted here, as they cancel out when forming the index.

To implement this construction, it is necessary to obtain price relatives that span multiple periods. I therefore construct multi-period predicted price relatives by chaining the

annual predicted price relatives from the hedonic model. In particular,

$$\begin{aligned}\frac{\widehat{p_{kt-j}}}{p_{kt}} &= \left(\prod_{n=t-j}^{t-1} \exp \left[\ln \left(\frac{p_{kn+1}}{p_{kn}} \right) \right] \right)^{-1}, \\ \frac{\widehat{p_{kt+i}}}{p_{kt}} &= \prod_{n=t}^{t+i-1} \exp \left[\ln \left(\frac{p_{kn+1}}{p_{kn}} \right) \right].\end{aligned}$$

These chained predicted price relatives provide the multi-period comparisons required to construct quality adjustment factors. Substituting these expressions into the previous definition yields

$$(5) \quad \widehat{\lambda}_{kt}^{t-j, t+i}(\alpha) = \left(\frac{\widehat{p_{kt+i}}}{p_{kt}} \right)^{\alpha} \left(\frac{\widehat{p_{kt-j}}}{p_{kt}} \right)^{1-\alpha} p_{kt}.$$

This construction ensures that quality-adjusted price changes align directly with the chained predicted price relatives implied by the hedonic model. For example,

$$\frac{p_{kt+1}/\widehat{\lambda}_{kt+1}^{t-j, t+i}(\alpha)}{p_{kt-1}/\widehat{\lambda}_{kt-1}^{t-j, t+i}(\alpha)} = \prod_{n=t-1}^t \exp \left[\ln \left(\frac{p_{kn+1}}{p_{kn}} \right) \right] = \frac{\widehat{p_{kt+1}}}{p_{kt-1}}$$

so that the resulting index corresponds to a full-imputation hedonic price index.

As highlighted in the discussion thus far, constructing quality adjustment factors requires predicted price relatives that span multiple periods within a given window. This creates a choice in how multi-period price relatives are constructed from the estimated model. The approach adopted in this paper chains adjacent predicted price relatives to form longer-horizon comparisons. An alternative approach is to estimate hedonic models directly over longer differences, producing predicted price relatives for multi-period comparisons without chaining. This alternative increases the number of observations contributing to the first-stage residuals used in the second stage of the EP-TV framework. However, using a slight abuse of notation, this approach implies a different notion of quality-adjusted price changes:

$$\frac{p_{kt+1}/\widehat{\lambda}_{kt+1}^{t-j, t+i}(\alpha)}{p_{kt-1}/\widehat{\lambda}_{kt-1}^{t-j, t+i}(\alpha)} = \frac{\left(\frac{\widehat{p_{kt+1}}}{p_{kt-j}} \right)^{\alpha} \left(\frac{\widehat{p_{kt+1}}}{p_{kt+i}} \right)^{1-\alpha}}{\left(\frac{\widehat{p_{kt-1}}}{p_{kt-j}} \right)^{\alpha} \left(\frac{\widehat{p_{kt-1}}}{p_{kt+i}} \right)^{1-\alpha}}.$$

This expression shows that, under the long-difference approach, quality-adjusted price

changes are no longer tied to adjacent-period changes, but instead reflect a combination of price relatives anchored at the endpoints of the window. As a result, intermediate price changes are inferred from long differences rather than chained intermediate price relatives. These alternative approaches to imputation reflect different tradeoffs between exploiting additional information when performing hedonics and preserving the interpretation of intermediate price movements. Exploring these alternatives in the construction of multilateral hedonic price indices is an interesting area for future research.

Figure 2 provides an empirical illustration of the quality adjustment factors constructed above by comparing observed and quality-adjusted price dynamics over the product life cycle. While both series exhibit upward-sloping profiles, the quality-adjusted price path increases more gradually. After ten years, observed prices exceed quality-adjusted prices by 6.6 percentage points. This gap reflects the contribution of changes in product characteristics to observed price growth. In particular, periodic redesigns introduce improvements in performance and features that raise observed prices even in the absence of pure price inflation. For example, the horsepower of a Ford F-Series truck increased by 2.5 times between 1980 and 2018, illustrating the magnitude of quality change embedded in observed prices.

Taken together, these patterns indicate that a portion of observed price growth over the product life cycle reflects improvements in product quality rather than increases in the cost of a constant-quality good. By expressing prices in a common, time-varying quality standard, the adjustment factors isolate this component and produce a measure of price change that more closely reflects inflation. As a result, indices constructed using hedonic imputation will generally exhibit lower measured inflation than indices based solely on observed prices.

3.5 Rolling Window Construction of Locally Transitive Indices

This subsection describes the construction of the locally transitive GQAUUV using a rolling window approach. I focus on the case where annual price indices are constructed without using information from future periods, yielding a timely and non-revisable measure of inflation. The index depends on several design choices, including the mixing parameter α and the maximum window length W , which are fixed throughout the construction. This procedure can be interpreted as the following sequence of steps:

- 1. Construct window-specific comparisons.** For each period t and each window length $w \in \{1, \dots, W\}$, construct time-varying quality adjustment factors using the procedure



Figure 2: Life Cycle of Prices

Notes: This figure plots average price dynamics over the life cycle of a product, indexed to one in the entry period. The observed price series reflects nominal prices, while the quality-adjusted series adjusts for changes in product characteristics using predicted price relatives from the hedonic model. The horizontal axis denotes years since product entry. Each point represents the average across products at a given age. Observations with a lifespan of over 15 years are excluded from this figure.

described in the previous subsection. These adjustment factors imply a set of bilateral log price changes $\Delta \ln P_{t-j,t-k}^{t-w,t}(\alpha)$ for all $0 \leq k < j \leq w$ and $w \in \{1, \dots, W\}$.

2. Stack comparisons within the window. Collect all implied log price changes into a vector $\Delta \ln P(\alpha, W)$, and construct the corresponding difference matrix A that maps log price levels to log price change.

3. Recover transitive price levels. Estimate log price levels by solving the least-squares problem:

$$\widehat{\ln P(\alpha)} = (\tilde{A}' \tilde{A})^{-1} \tilde{A}' \Delta \ln P(\alpha, W),$$

where $\tilde{A}$ denotes the difference matrix from equation (4) with the column corresponding to period $t - W$ removed, imposing the normalization $\widehat{\ln P_{t-W}(\alpha)} = 0$.

4. Calculate inflation rates. The implied inflation rate between periods $t - 1$ and t is then given by

$$\widehat{\Delta \ln P}_{t-1,t}(\alpha) = \exp \left(\widehat{\ln P}_t(\alpha) - \widehat{\ln P}_{t-1}(\alpha) \right).$$

5. Roll the window forward. Advance the window to $\{t - W + 1, \dots, t + 1\}$, update the set of comparisons, and repeat Steps 1–3. Chaining the resulting adjacent-period inflation rates yields the full price index.¹¹

This procedure combines multiple overlapping comparisons into a single transitive price index by combining them into a set of price levels that is consistent with the available comparisons. Because comparisons are constructed over different window lengths, the resulting index incorporates information from both short- and long-horizon price movements while remaining responsive to newly available data.

3.6 Choice of Maximum Window Length

To implement the rolling-window construction of the locally transitive GQAUUV, I must select a maximum window length W . This choice governs the horizon over which local transitivity is imposed and determines how far quality adjustment is allowed to span product cycles. In addition, the choice of W determines the amount of data required before the index can be constructed. Because the rolling procedure uses information from the previous W periods, the first W observations in the sample cannot be used to compute inflation rates. As a result, larger values of W increase the initial sample period that is effectively unavailable for estimation.

To guide the selection of W , I align the window length with the empirical life cycle of products in the data. In the US automobile market, vehicles undergo frequent entry, exit, and redesigns, implying that comparisons over long horizons may involve goods that are not economically comparable. Choosing a window length that corresponds to the typical product life cycle ensures that quality adjustment is performed over economically meaningful comparisons, while limiting the influence of long-run changes in product characteristics that may not be fully captured by the hedonic model.

Figure 3 presents the cumulative distribution of product lifespans following initial entry. The unadjusted series measures the number of years a vehicle model is observed in the data, while the design-adjusted series treats major redesigns as new product entries. Both series exhibit substantial product turnover, with a large fraction of vehicles present

¹¹de Haan (2015b) studies the impact that splicing methods have on the estimated rate of inflation for multilateral price indices. In the set-up provided in this section, I choose to use a window splice. A mean splice can be used as an alternative and can be utilized in future research.

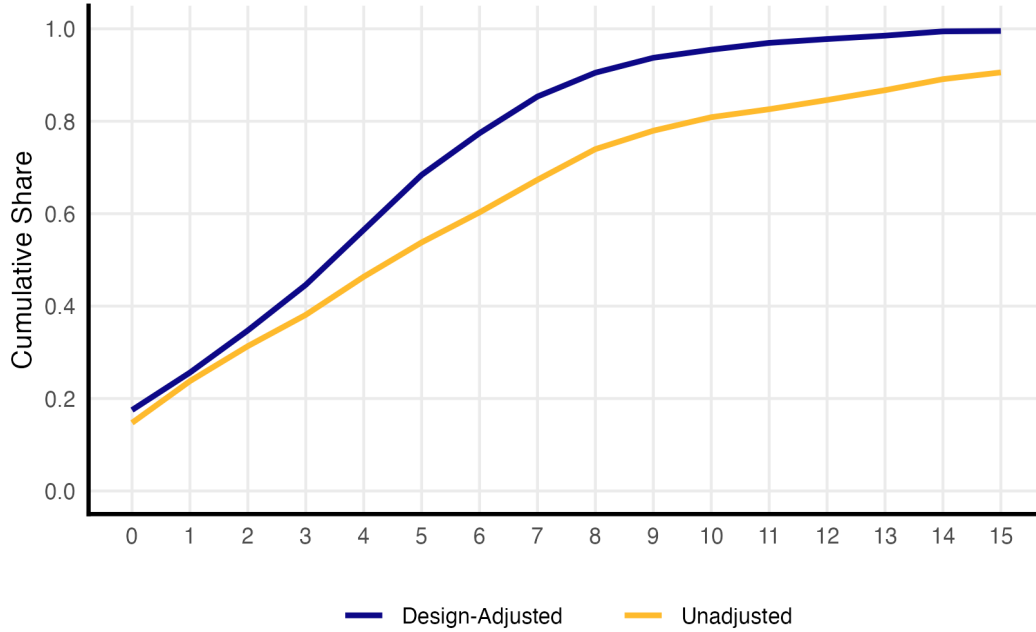


Figure 3: Cumulative Distribution of Model Age at Time of Exit

Notes: This figure reports the cumulative distribution of product lifespans in the sample. The unadjusted series measures the number of years a vehicle is observed following entry, while the design-adjusted series measure the number of years a vehicle-design is observed following entry. Observations with a lifespan of over 15 years are excluded from this figure.

for only a short period and a majority exiting within five years. The design-adjusted series converges to one more rapidly, reflecting the frequency of redesigns even when model names persist over time.

Motivated by these patterns, I select a maximum window length of $W = 7$ years in the main empirical analysis. At this horizon, approximately 67 percent of vehicles and 85 percent of vehicle-designs have exited the sample. This choice balances the gains from imposing local transitivity over a sufficiently long horizon with the need to maintain comparisons among economically similar goods. To assess the robustness of this choice, I also construct indices using a longer window length of up to $W = 10$ years and an alternative index that allows the maximum window length to expand over time.

Finally, the choice of W can also be selected to account for calendar-based sources of price changes. In the construction of multilateral indices such as RYGEKS, the window length is often chosen to ensure that comparisons span a full cycle of seasonal or calendar-driven price changes, thereby reducing sensitivity to temporary fluctuations, which is a key source of chain drift in empirical setting using high-frequency point-of-sale data. Although the empirical analysis in this paper is conducted using annual data, the same

logic applies more generally to other contexts. Selecting a sufficiently long window helps average over short-run fluctuations in prices while preserving the responsiveness of the index to changes in product quality over time.

4 Empirical Results

This section evaluates the empirical implications of the locally transitive GQAUV for the US automobile market. I construct annual price indices from 1987 to 2018 and focus on how the choice of window length governs the tradeoff between maintaining transitivity and incorporating new information on product quality. The preferred specification for the locally transitive GQAUV uses a maximum window length of seven years and a mixing parameter of 0.5.

I begin by examining how the locally transitive GQAUV varies with the choice of window length and mixing parameter, highlighting the role these parameters play in determining the persistence of the quality standards used to measure inflation. I then present product-level evidence to illustrate why longer windows yield higher measured inflation by slowing the incorporation of quality improvements. With this mechanic in place, I compare the locally transitive GQAUV to a range of alternative indices, including bilateral and multilateral Tornqvist indices that differ in their use of observed versus predicted price relatives, chaining structure, and aggregation method. I conclude by assessing the robustness of the results to alternative design choices, including the set of local comparisons used in aggregation and weighting schemes.

4.1 Mechanics of the Locally Transitive GQAUV

This subsection examines the mechanics of the locally transitive GQAUV and illustrates how its key design parameters govern the tradeoff between maintaining transitivity and incorporating new information on product quality. In particular, I first focus on the role of the maximum window length and the mixing parameter in determining how long the quality standard is held fixed and how quickly it evolves over time. I then provide product-level evidence demonstrating that this pattern arises because longer windows slow the rate at which improvements in product characteristics are incorporated into the benchmark used to construct quality-adjusted prices. Together, these results establish the mechanism through which the locally transitive GQAUV differs from alternative index constructions.

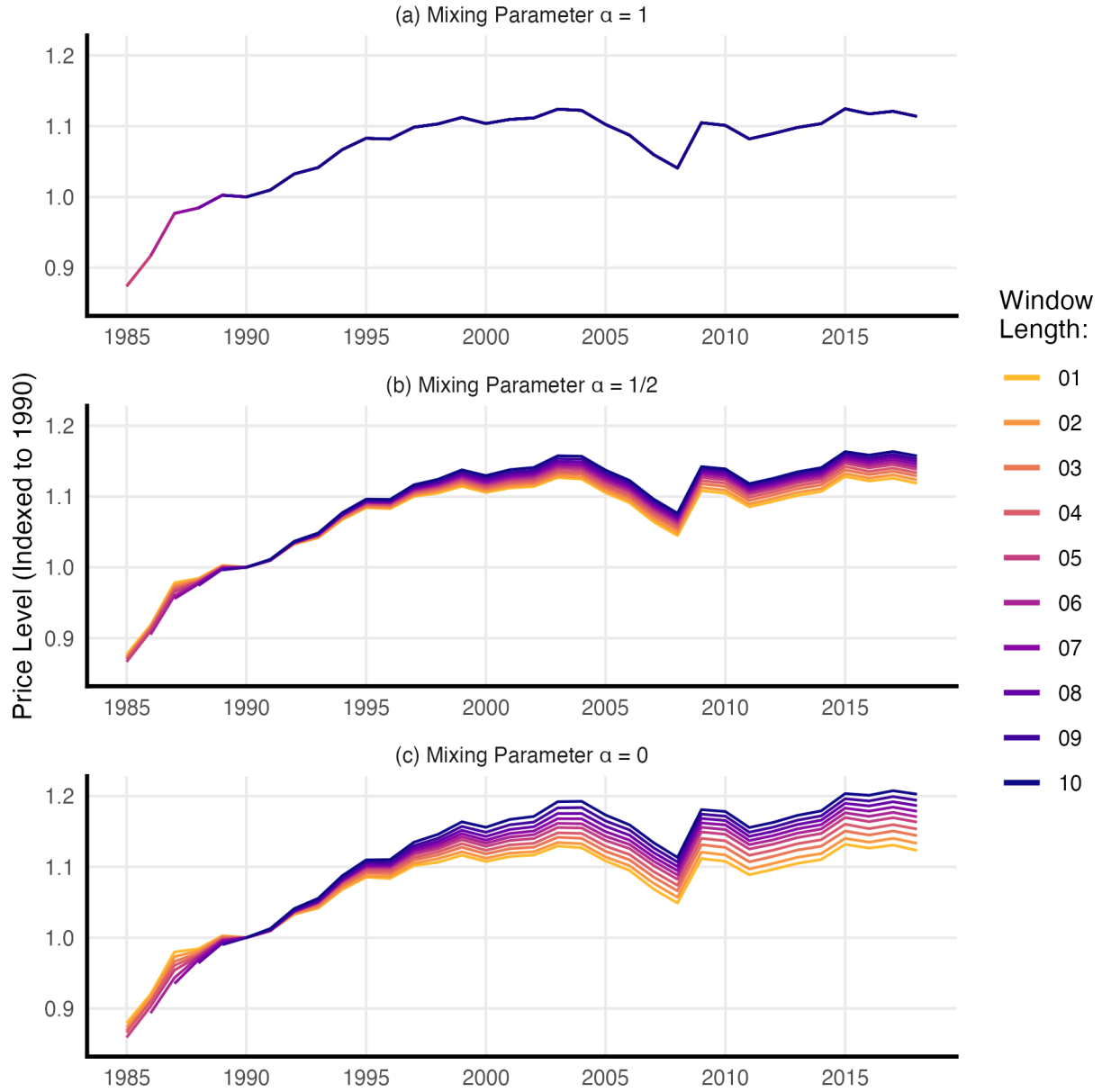


Figure 4: Varying Window Length and Mixing Parameter

Notes: This figure plots the locally transitive GQAUV under alternative choices of the mixing parameter α and the maximum window length. Each panel corresponds to a different value of α , while the colored lines represent indices constructed using window lengths ranging from one to ten years.

4.1.1 Choice of Window Length and Mixing Parameter

Figure 4 illustrates how the locally transitive GQAUV varies with respect to the maximum window length and the choice of the mixing parameter α . Panel (a) shows that when $\alpha = 1$, the resulting index is largely invariant to the choice of window length. In this

case, the quality standard is defined entirely using current-period information, so that extending the comparison window does not alter the constructed price relatives. Under this specification, the resulting index reflects a construction that places full weight on current-period information in defining the quality standard, similar in spirit to a Laspeyres-type index.

At the opposite extreme, panel (c) shows that when $\alpha = 0$, the resulting index depends strongly on the maximum window length. In this case, the quality standard is defined entirely using information from earlier periods, so that extending the window changes how goods are evaluated across time. As a result, longer windows produce systematically higher measured price levels.

Panel (b) presents the intermediate case with $\alpha = 0.5$, which balances these two sources of information. Under this specification, the resulting series is less sensitive to the choice of window length while still incorporating information from each available window. When the window length is equal to one and $\alpha = 0.5$, the resulting construction reduces to adjacent-period comparisons with symmetric weighting across periods, analogous to a Tornqvist-type index. As the window length increases, the resulting index reflects comparisons constructed over longer horizons, which stabilize the underlying quality standard across periods.

These results reflect the conceptual framework developed in the earlier section. Differences across index number formulas arise from how the underlying quality standard is defined and how predicted price relatives are aggregated across periods. The mixing parameter α governs how information from different periods enters the construction of the quality standard, while the window length determines the temporal scope over which comparisons are formed. The sensitivity of the resulting index to these parameters highlights the role of these modeling choices in shaping measured price changes and clarifies how the GQAUV nests commonly used bilateral and multilateral indices as special cases of the broader construction.

4.1.2 Product-Level Evidence

Next, I examine how the choice of window length affects quality-adjusted prices at the item-level. Figure 5 shows that extending the window used to construct quality adjustment factors systematically increases the level of quality-adjusted prices within vehicle designs. This pattern provides direct evidence that the window length governs how quickly improvements in product characteristics are absorbed into the underlying quality standard, rather than reflecting misspecification in the estimation of characteristic-

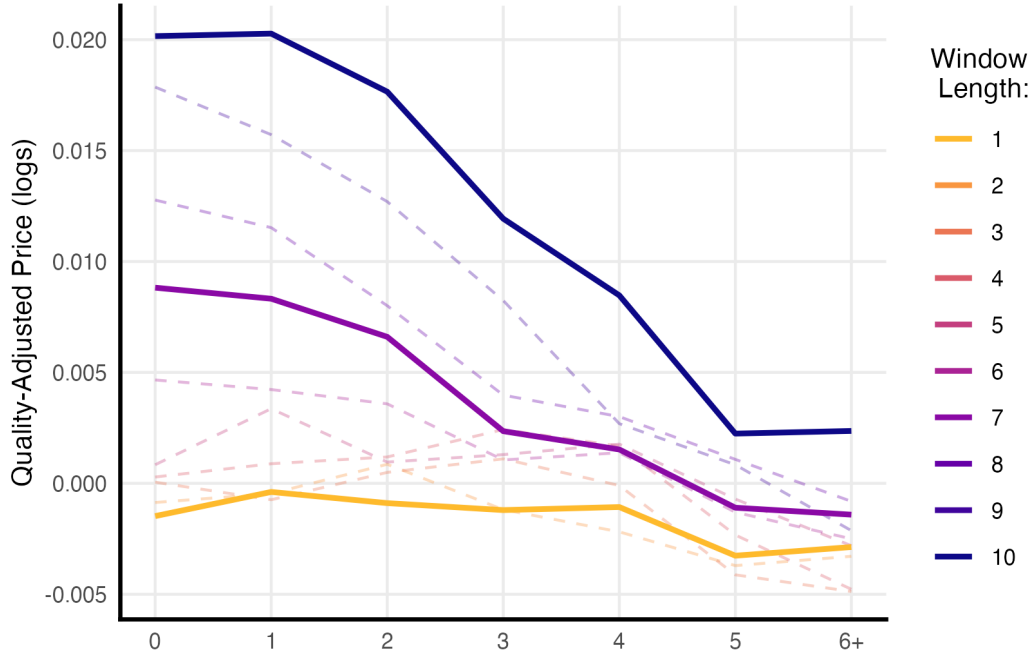


Figure 5: Effect of Window Length on Quality-Adjusted Prices

Notes: This figure plots the median log quality-adjusted price across vehicle-designs by years since a vehicle's model redesign. For each vehicle-design, observations are aligned relative to the year of a redesign (year 0), and the first generation of each vehicle is excluded. Quality-adjusted prices are constructed using predicted price relatives from the hedonic model and vary according to the window length used to define underlying quality adjustment factors. Each value represents the median log quality-adjusted price across vehicle-designs at a given age and window length. Observations are restricted to the sample of vehicles present after 1990 to accommodate all window lengths.

constant prices.

A common benchmark in hedonic price measurement is that, under a correctly specified model with complete information, quality-adjusted prices would remain constant following a product redesign, as changes in characteristics would be fully accounted for. However, this property arises under a specific updating rule for the quality standard and does not uniquely characterize a quality-adjusted price index.

As emphasized in the conceptual framework, a quality-adjusted price index requires a quality standard, and the evolution of this standard depends on how new information is incorporated over time. Bilateral hedonic indices update the quality standard each period using contemporaneous characteristic-constant price changes. This allows the benchmark to adjust immediately to improvements in product characteristics, yielding relatively flat quality-adjusted price profiles over the product life cycle.

In contrast, longer windows impose local transitivity across multiple periods, constraining how the quality standard evolves within the window. This constraint slows the

rate at which improvements in product characteristics are incorporated into the benchmark. As a result, cumulative quality improvements are not immediately absorbed and instead appear as increases in quality-adjusted prices.

The divergence across window lengths therefore reflects differences in the speed at which quality growth is incorporated into the reference standard rather than measurement error. Short windows produce a rapidly evolving benchmark that attributes most changes to quality, while longer windows impose a more stable standard that attributes a larger share of observed price changes to quality-adjusted prices.

Importantly, this pattern does not reflect a failure to account for quality change with hedonics, but instead arises from differences in how quality standards are defined across bilateral comparisons. Indices such as GEKS aggregate bilateral comparisons constructed using time-varying quality standards, allowing the benchmark to adjust each period. In contrast, the locally transitive GQAUUV places greater weight on comparisons within windows where the quality standard is held fixed. This constraint limits successive redefinitions of the benchmark and increases the number of bilateral comparisons constructed under a common quality standard, thereby yielding higher measured quality-adjusted prices in the aggregate.

4.2 Comparison to Alternative Price Indices

This subsection compares the estimated rate of inflation calculated using the locally transitive GQAUUV to alternative index formulas. I first benchmark the GQAUUV against chained indices to illustrate how alternative approaches to quality adjustment and transitivity affect measured inflation. I then compare the GQAUUV to a direct hedonic Tornqvist index, which provides a fixed-base benchmark, to assess whether stabilizing the evolution of quality standards yields a measure of inflation consistent with direct comparisons. Appendix A provides a complete description of alternative indices included in the analysis.

4.2.1 Chained Bilateral Indices

Figure 6 compares the locally transitive GQAUUV to several benchmark price indices. A central result is that full-imputation hedonic Tornqvist produces substantially lower measured inflation, while the locally transitive GQAUUV yields higher inflation by stabilizing quality standards within windows rather than updating them at each comparison. The locally transitive GQAUUV lies above the hedonic Tornqvist index, but below the standard chained Tornqvist index.

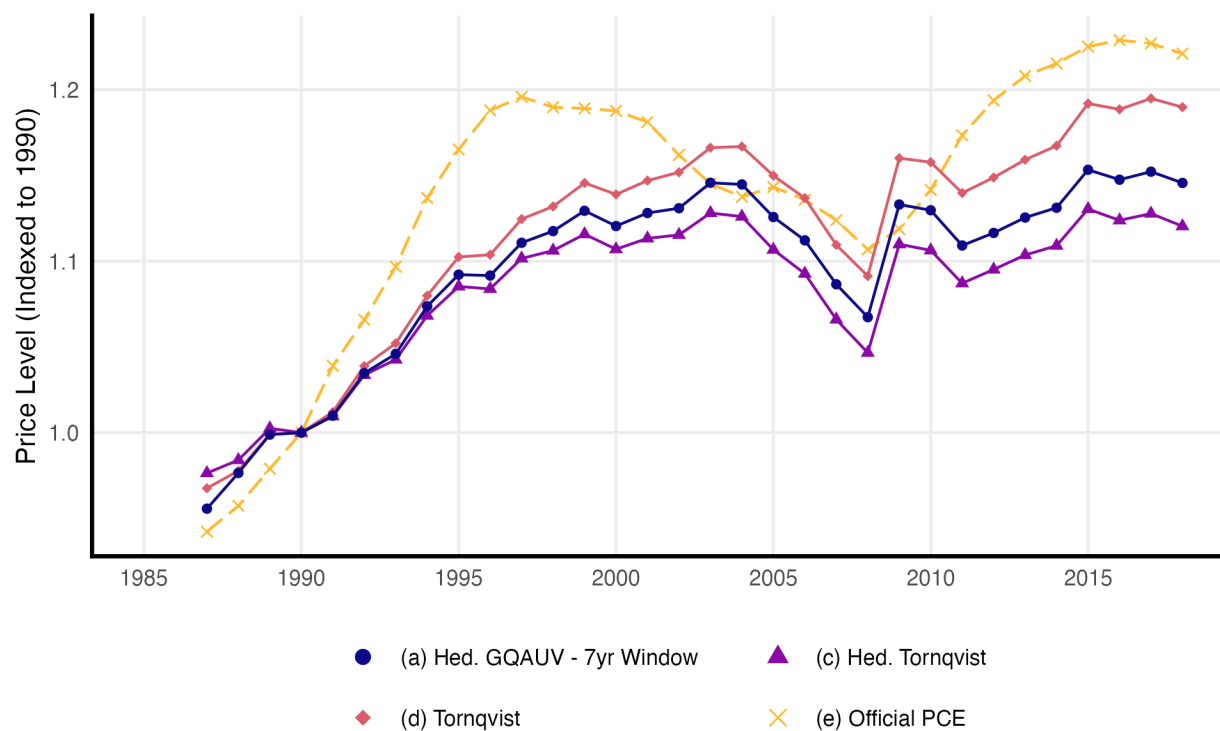


Figure 6: Cumulative Inflation compared to Chained Bilateral Price Indices

Notes: This figure plots cumulative price indices for new automobiles constructed using alternative methods. The locally transitive GQAUV is computed using a maximum window length of seven years and a mixing parameter of 0.5. The standard chained Tornqvist index is based on observed prices, while the hedonic chained Tornqvist index uses predicted price relatives from the hedonic model. The official PCE price index for “New motor vehicles (55)” is shown for comparison.

To provide context, the official PCE price index for new automobiles tracks the standard chained Tornqvist index, suggesting that both series reflect similar underlying price movements in the raw data. Incorporating hedonic adjustments lowers measured inflation relative to the standard chained index, as shown by the gap between the standard and hedonic Tornqvist indices. This difference reflects how frequently quality standards are updated. The hedonic Tornqvist index incorporates new information about product characteristics at each comparison, attributing a larger share of observed price changes to improvements in product quality.

Table 3 summarizes these price changes, reporting average annual inflation rates and cumulative price changes between 1987 and 2018 for the same set of indices. Hedonic price indices estimate a substantially lower rate of inflation relative to the standard chained Tornqvist index. The standard chained Tornqvist index implies an average annual inflation rate of 0.669 percent, compared to 0.445 percent for the hedonic Tornqvist.

Table 3: Annual Rate of Inflation compared to Chained Bilateral Price Indices

	Percent Change (1)	Cumulative Change (2)
Hedonic GQAUV - 7yr Window	0.584	19.8
Hedonic Tornqvist	0.445	14.7
Tornqvist	0.669	23.0

Notes: This table reports average annual inflation rates and cumulative price changes for new automobiles from 1987 to 2018. The locally transitive GQAUV is constructed using a maximum window length of seven years and a mixing parameter of 0.5. Hedonic indices use predicted price relatives from the hedonic model. All cumulative changes are computed relative to 1987.

These differences translate into a gap of roughly 8 percentage points in cumulative price growth over the sample period.

Relative to these benchmarks, the GQAUV's average annual inflation rate is 0.584 percent, approximately 31 percent higher than the hedonic Tornqvist estimate, and corresponds to an additional 5.1 percentage points of cumulative price growth over the sample period. These magnitudes indicate that alternative choices about how quality standards evolve and how transitivity is imposed can lead to materially different conclusions about long-run inflation.

Taken together, these results illustrate how chained bilateral indices reflect a particular balance between updating quality standards and maintaining transitivity. By updating quality standards at each comparison, the hedonic Tornqvist index attributes a larger share of price changes to quality improvements, resulting in lower measured inflation. By contrast, the locally transitive GQAUV stabilizes quality standards within windows before aggregation, yielding a higher measured rate of inflation. These findings demonstrate that differences across indices arise from how quality standards are updated and how transitivity is implemented in practice.

4.2.2 Multilateral Indices

Table 4 compares average price trends from the the locally transitive GQAUV to alternative multilateral indices. A clear empirical pattern emerges: indices that impose transitivity over longer horizons and therefore maintain a more persistent quality standard produce higher measured inflation than those that update quality standards more frequently. In particular, the locally transitive GQAUV yields average annual inflation of 0.584 percent, closely aligning with the full-sample time-product-dummy index (0.582

Table 4: Annual Rate of Inflation across Alternative Multilateral Indices

	Percent Change (1)	Cumulative Change (2)
Hedonic GQAUV - 7yr Window	0.584	19.8
Time-Product-Dummy - Full-Sample	0.582	19.7
Rolling Time-Dummy - 7yr Window	0.462	15.3
Hedonic Rolling GEKS - 7yr Window	0.464	15.4

Notes: This table reports average annual inflation rates and cumulative price changes for new automobiles from 1987 to 2018. The locally transitive GQAUV is computed using a maximum window length of seven years and a mixing parameter of 0.5.

percent), while the rolling time-dummy and hedonic GEKS indices produce substantially lower estimates of approximately 0.46 percent.¹² These differences can be understood through how each method balances the persistence of transitivity against the timeliness of quality adjustment.

Figure 7 complements these results by illustrating how price dynamics differ across indices. While the locally transitive GQAUV and the full-sample time-product-dummy index imply nearly identical average inflation rates, they follow distinct dynamic paths. In particular, the time-product-dummy index evolves more rapidly in the early part of the sample and then more smoothly thereafter. The rolling time-dummy index exhibits similar dynamics, but its price level remains systematically lower throughout the sample.

Taken together, these results suggest that while all multilateral price indices are transitive, they differ empirically according to how transitivity is enforced over time and how quality adjustment is applied. Methods that stabilize the quality standard over longer horizons attribute a larger share of observed price changes to inflation, while those that update the standard more frequently attribute a greater share to quality improvement.

A key advantage of this comparison is that the hedonic Tornqvist, RGEKS, and locally transitive GQAUV indices are constructed from a common set of quality-adjusted price relatives derived from the EP-TV framework. As a result, differences across these indices isolate the role of the aggregation procedure itself. In contrast, comparisons involving time-dummy and time-product-dummy indices may reflect not only differences in how transitivity is imposed, but also differences in how product characteristics are incorporated through regression-based normalization.

¹²To motivate its inclusion, [Aizcorbe and Ripperger-Suhler \(2026\)](#) use a full-sample time-product-dummy as a benchmark index to measure chain drift among other indices including the RYGEKS index.

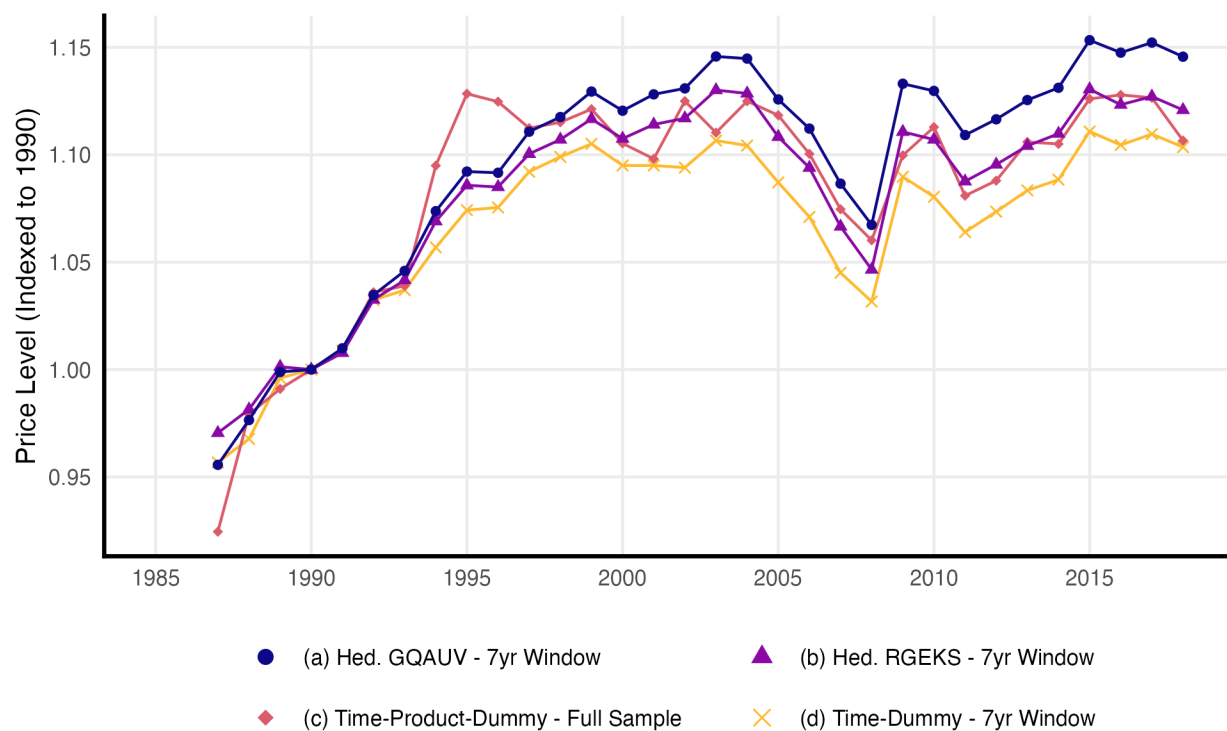


Figure 7: Comparisons to Alternative Multilateral Indices

Notes: This table reports average annual inflation rates and cumulative price changes for new automobiles from 1987 to 2018. The locally transitive GQAUV is computed using a maximum window length of seven years and a mixing parameter of 0.5.

The locally transitive GQAUV therefore provides a transparent way to navigate this tradeoff, delivering estimates that closely match fully transitive approaches in long-run magnitude while preserving a controlled degree of timeliness in the evolution of the quality standard.

4.2.3 Direct Indices

To evaluate whether the locally transitive GQAUV recovers a precise measure of average quality-adjusted price changes, Figure 8 plots it against a direct hedonic Tornqvist. Unlike chained indices, direct indices compare prices across all periods relative to a fixed base year and therefore do not suffer from chain drift. As a result, they provide a natural benchmark for assessing whether alternative approaches preserve a consistent quality standard over time.

A limitation of direct indices, however, is that they depend on the choice of base year, as each base year depends on different set of vehicles with positive sales. To address this

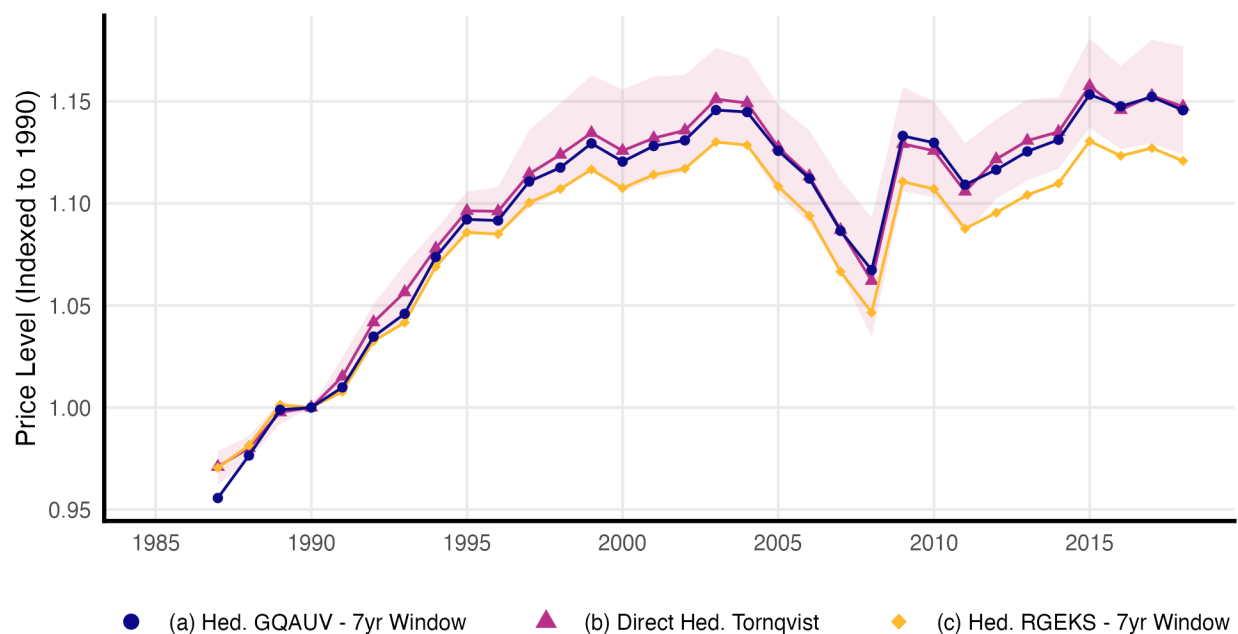


Figure 8: Comparisons to Direct Indices

Notes: This figure compares the locally transitive GQAUV and hedonic RGEKS indices to a direct full-imputation hedonic Tornqvist index. A direct index is constructed separately for each base year in the sample, and the reported series corresponds to the geometric mean across base years. The shaded region denotes the range of values across these base-year-specific indices.

issue, I construct a family of direct hedonic Tornqvist indices using each year in the sample as the base period. I then aggregate these indices by taking the geometric mean across base years to obtain a single representative direct index that is unbiased to any potential outliers. Lastly, the shaded region in Figure 8 represents the range of direct indices across base years, capturing the sensitivity of the direct index to the choice of reference period. This range reflects differences in product availability and quality standards across base years.

Relative to this benchmark, the locally transitive GQAUV closely tracks the average direct index, while the hedonic RGEKS index lies systematically below it. This pattern reflects differences in how the two indices balance the tradeoff between maintaining transitivity and updating quality standards over time. The GQAUV enforces transitivity within windows by holding quality standards fixed locally, which yields comparisons that are more closely aligned with fixed-base measures of inflation. By contrast, the RGEKS index enforces transitivity at the aggregation stage while allowing quality standards to update at each bilateral comparison, resulting in a greater influence of time-varying reference standards on the measured rate of inflation. Taken together, these results illustrate how

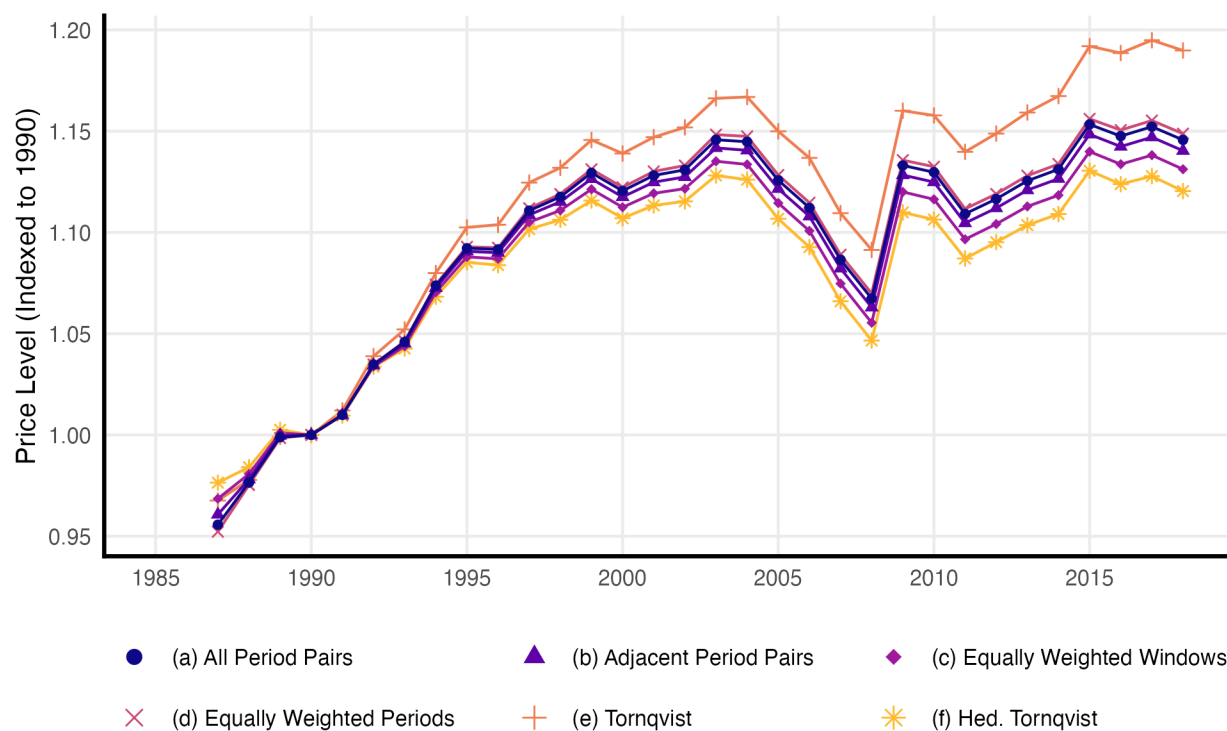


Figure 9: Varying Local Comparisons and Weighting Scheme

Notes: This figure compares alternative constructions of the locally transitive GQAUUV that vary the set of bilateral inputs and the weighting scheme used in aggregation. The baseline specification uses all bilateral comparisons within each window with equal weights. Alternative specifications restrict attention to adjacent-period comparisons, assign equal weight to each window, or assign equal weight to each bilateral comparison. The standard and hedonic Tornqvist indices are included for comparison.

alternative index constructions correspond to different choices along this tradeoff, with the GQAUUV placing greater weight on maintaining consistent quality standards and the RGEKS index implicitly placing greater weight on incorporating new information about product quality.

4.3 Robustness

Figure 9 examines how the locally transitive GQAUUV varies with alternative choices in the second-stage aggregation, including both the set of bilateral inputs and the weighting scheme applied to those inputs. The preferred specification uses all bilateral comparisons within a given window and assigns equal weight to each comparison.

Because the GQAUUV is locally transitive within windows, the index can alternatively be constructed using only adjacent-period comparisons. This restriction reduces the set of bilateral inputs by placing less weight on longer-horizon comparisons that arise as the

window length increases. As shown in the figure, this yields a similar but slightly lower price level, reflecting the diminished role of comparisons constructed under a common quality standard over longer horizons.

The remaining specifications vary the weighting of bilateral comparisons. Series (c) assigns equal weight to each window, while series (d) assigns equal weight to each bilateral comparison across periods. Reweighting toward shorter windows places greater emphasis on more recent comparisons and produces an index that lies closer to specifications that update quality standards more frequently. Conversely, reweighting toward longer windows increases the influence of comparisons constructed under a fixed quality standard, yielding slightly higher measured price levels.

Taken together, these results show that the level of the GQAUV varies modestly with the choice of inputs and weights, but the underlying mechanism remains unchanged. Specifications that place greater weight on comparisons constructed under a fixed quality standard consistently produce higher measured prices than those that rely more heavily on frequently updated comparisons.

This exercise therefore highlights that differences in measured inflation are not driven by specific aggregation choices, but instead reflect how the underlying quality standard is defined and maintained across comparisons. By stabilizing this standard within bilateral comparisons prior to aggregation, the locally transitive GQAUV exhibits the robust tradeoff between timely quality adjustment and the persistence of local transitivity.

5 Conclusion

Measuring inflation in the presence of product turnover and evolving quality requires balancing two competing objectives: incorporating new information in a timely manner while maintaining a consistent standard for comparing products over time. Standard approaches based on chained bilateral indices achieve timeliness but implicitly redefine the quality standard in each comparison, introducing path dependence in measured price changes. Common multilateral methods address this issue by enforcing transitivity at the second-stage of aggregation stage, but continue to rely on bilateral inputs that embed inconsistent quality standards.

This paper develops a new approach to price measurement that reinterprets matched-model indices through the lens of quality-adjusted unit value indices. I show that commonly used geometric indices can be understood as special cases of a broader class of indices that differ in how they define and update the underlying quality standard. Build-

ing on this insight, I introduce a locally transitive geometric quality-adjusted unit value index (GQAUV) that anchors quality adjustment within overlapping windows of time. This construction allows the quality standard to evolve gradually while remaining consistent within each window, providing a middle ground between fixed-base and fully chained approaches.

Applying this framework to the US automobile market yields systematically higher measured inflation as transitivity is enforced over a longer time horizon. Moreover, the locally transitive GQAUV produces higher cumulative price growth than both hedonic chained and rolling multilateral indices, with differences that are economically meaningful and increase over time. These results reflect the central mechanism of the paper: when quality standards are repeatedly redefined in chained comparisons, improvements in product characteristics are more readily absorbed into the benchmark used to evaluate prices, leading to lower measured inflation. By stabilizing the quality standard within windows, the locally transitive GQAUV attributes a larger share of observed price changes to price growth rather than to changes in the reference standard.

More broadly, these findings highlight that differences across price indices arise not only from the use of hedonic methods or multilateral aggregation, but from how quality standards are defined and updated over time. This perspective has important implications for official statistics, particularly as statistical agencies increasingly rely on high-frequency and high-dimensional data sources. In such environments, where product turnover is rapid and product characteristics evolve continuously, the choice of how to anchor quality adjustment may have first-order effects on measured inflation.

Taken together, the locally transitive GQAUV provides a flexible and interpretable framework for price measurement that reconciles timeliness and consistency in the presence of evolving product quality.

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A Description of Alternative Indices

Official PCE. Data for the Personal Consumption Expenditures (PCE) price index are obtained from the Bureau of Economic Analysis (BEA), available at the official BEA website. I use Table U20404-A, published on March 27, 2025, which reports annual price indices. The series used in the analysis corresponds to “New motor vehicles (55)” (series DNMVRG). This series is used as an external benchmark and is not constructed from the underlying automobile microdata. For comparability with the automobile price indices constructed in this paper, the series is normalized to a common base year and restricted to the sample period 1987–2018.

Chained standard Tornqvist. The chained Tornqvist index is constructed by linking bilateral Tornqvist indices across adjacent periods. For periods $t - 1$ and t , the bilateral Tornqvist index is given by:

$$\text{Tornqvist}_{t-1,t} = \prod_{k \in C_{t-1,t}} \left(\frac{p_{kt}}{p_{kt-1}} \right)^{\frac{1}{2}(s_{kt-1} + s_{kt})},$$

where $C_{t-1,t}$ denotes the set of products observed in both periods and s_{kt} denotes the expenditure share of product k in period t , defined as

$$s_{kt} = \frac{p_{kt}q_{kt}}{\sum_j p_{jt}q_{jt}}.$$

The chained Tornqvist index is then constructed recursively as

$$P_t = P_{t-1} \times \text{Tornqvist}_{t-1,t},$$

and normalized such that $P_{1990} = 1$.

Chained Full-Imputation Hedonic Tornqvist. The chained full-imputation hedonic Tornqvist index is constructed by linking bilateral hedonic Tornqvist indices across adjacent periods. For periods $t - 1$ and t , the bilateral index is given by:

$$\text{Hed. Tornqvist}_{t-1,t} = \prod_{k \in \Omega_{t-1} \cup \Omega_t} \left(\exp \left(\widehat{\Delta \ln p_{kt}} \right) \right)^{\frac{1}{2}(s_{kt-1} + s_{kt})},$$

where Ω_t denotes the set of products observed in period t , and $\widehat{\Delta \ln p_{kt}}$ denotes the predicted log price change for product k between periods $t - 1$ and t , obtained from the EP-TV hedonic model described in Section 3.2. Because the data form a strongly balanced panel, products with zero sales remain in the sample and are assigned zero expenditure share in the corresponding period. The expenditure share s_{kt} is defined as

$$s_{kt} = \frac{p_{kt}q_{kt}}{\sum_{j \in \Omega_t} p_{jt}q_{jt}}.$$

The chained full-imputation hedonic Tornqvist index is then constructed recursively as

$$P_t = P_{t-1} \times \text{Hed. Tornqvist}_{t-1,t},$$

and normalized such that $P_{1990} = 1$.

Hedonic rolling GEKS (7-year window). The index is constructed by applying the GEKS procedure over a rolling window of length $W = 7$ and linking adjacent-period movements using a window splice. For each period t , GEKS levels are computed within the window $\{t - W + 1, \dots, t\}$, and the implied adjacent-period movement between $t - 1$ and t is recovered from that window.

The inputs to the GEKS index are multi-period price relatives constructed from the full-imputation hedonic Tornqvist index. For periods $t - j$ and t , these are defined as:

$$\frac{\mathbb{P}_t}{\mathbb{P}_{t-j}} = \prod_{k \in \Omega_{t-j} \cup \Omega_t} \left[\prod_{i=1}^j \exp(\widehat{\Delta \ln p_{k,t-j+i}}) \right]^{\frac{1}{2}(s_{k,t-j} + s_{kt})},$$

where $\widehat{\Delta \ln p_{k,t}}$ denotes the predicted log price change for product k between periods $t - 1$ and t , obtained from the EP-TV hedonic model.

Using these inputs, the GEKS index over a window of length W is constructed as:

$$\text{GEKS}_{t-1,t}^{(t)} = \prod_{w=1}^W \left[\frac{\mathbb{P}_{t-w}}{\mathbb{P}_{t-1}} \cdot \frac{\mathbb{P}_t}{\mathbb{P}_{t-w}} \right]^{\frac{1}{W}},$$

where all terms are evaluated using data within the window ending at period t . Because the window is rolled forward one period at a time, consecutive estimates are constructed using overlapping sets of reference periods. The window splice defines the final adjacent-

period index as:

$$P_t = P_{t-1} \times \frac{\text{GEKS}_t^{(t-W+1,t)}}{\text{GEKS}_{t-1}^{(t-W+1,t)}}.$$

The resulting series is normalized such that $P_{1990} = 1$.

Rolling time-dummy index (7-year window). The rolling time-dummy index is constructed by estimating a weighted time-dummy hedonic regression over rolling windows of length seven years. For each window beginning in period b and ending in period $b + 6$, I estimate

$$\ln p_{kt} = \sum_{\tau=b+1}^{b+6} \delta_{\tau} \mathbf{1}\{t = \tau\} + X'_{kt} \beta + \gamma_k + u_{kt},$$

where p_{kt} denotes the price of product k in period t , X_{kt} includes observed product characteristics, γ_k denotes an item fixed effect, and observations are weighted by expenditure shares within each year. Because product characteristics can vary across years, I average characteristic values within the given window of time, where binary indicators are equal to one if that average value is non-zero.

To obtain an adjacent-period inflation rate on a rolling basis, I use two overlapping regression windows. For a given terminal year $t = b + 6$, I estimate the time-dummy regression over the window $[b, b + 6]$ and recover the coefficient $\delta_t^{(b)}$ on the terminal year. I also estimate the regression over the overlapping window $[b - 1, b + 5]$, which yields coefficients $\delta_{t-1}^{(b-1)}$ and $\delta_{t-6}^{(b-1)}$. The implied log price change between periods $t - 1$ and t is then constructed as

$$\widehat{\Delta \ln P_t} = \delta_t^{(b)} - (\delta_{t-1}^{(b-1)} - \delta_{t-6}^{(b-1)}).$$

This expression aligns the terminal-year coefficient from the current window with the corresponding base-period normalization from the overlapping window, yielding a consistent estimate of the adjacent-period price change. The full price index is then constructed by cumulating these adjacent-period log changes and normalized such that $P_{1990} = 1$.

Full-sample time-product-dummy index. The full-sample time-product-dummy index is constructed by estimating a single weighted regression over the full sample (1980-2018):

$$\ln p_{kt} = \sum_{\tau \neq 1990} \delta_{\tau} \mathbf{1}\{t = \tau\} + \gamma_k + u_{kt},$$

where γ_k denotes an item fixed effect and observations are weighted by expenditure shares within each year. Unlike the rolling time-dummy specification, this regression

does not include observed product characteristics, and product heterogeneity is absorbed entirely by the item fixed effects. The omitted year is 1990, so that each estimated time dummy coefficient measures the log price level in year τ relative to 1990 for a constant-quality product. The corresponding price index is obtained as

$$P_t = \exp(\delta_t),$$

with $\delta_{1990} = 0$.

Direct full-imputation hedonic Tornqvist. The direct full-imputation hedonic Tornqvist index is constructed separately for each base year $b \in \{1987, \dots, 2018\}$. Let $\hat{p}_{kt}$ denote the hedonic price for product k in period t , obtained recursively by chaining adjacent-period predicted price relatives from the EP-TV hedonic model. For a given base year b , the direct index for period t is defined as

$$\text{Direct Hed. Tornqvist}_{b,t} = \exp \left(\sum_{k \in \Omega_b \cup \Omega_t} \frac{1}{2} (s_{kb} + s_{kt}) [\ln \hat{p}_{kt} - \ln \hat{p}_{kb}] \right),$$

where s_{kt} denotes the expenditure share of product k in period t . Because the panel is strongly balanced, products are present in all periods, with zero expenditure share assigned when sales are zero. For each base year b , the resulting index is normalized such that

$$\text{Direct Hed. Tornqvist}_{b,1990} = 1.$$

Because the direct index depends on the choice of base year, I construct this index separately for each base year and then aggregate across base years. The reported series is the geometric mean across base-year-specific indices:

$$P_t = \exp \left(\frac{1}{B} \sum_{b=1}^B \ln (\text{Direct Hed. Tornqvist}_{b,t}) \right),$$

where B denotes the number of base years. I also report the minimum and maximum values across base-year-specific indices to illustrate sensitivity to the choice of reference period.